

Enhancing Digital Government and Economy

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Assignment Name: Design Test Cases for Facebook Create New Account and Login

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Title: Design Test Cases for Facebook Create New Account and Login

Objective:

The objectives of this assignment are:

1. To understand the functional requirements of **Facebook's Create New Account** and **Login** features.
2. To design comprehensive **test cases** covering positive scenarios, negative scenarios, boundary values, and edge cases.
3. To apply **black-box testing techniques** (equivalence partitioning, boundary value analysis, error guessing) to real-world web application features.
4. To include **non-functional test cases** such as security, performance, usability, and compatibility.
5. To organize test cases in a professional, repeatable format suitable for execution by an SQA team.
6. To demonstrate how structured test case design improves software quality and defect detection.

Methodology:

The following methodology was followed to design test cases for Facebook's Create New Account and Login functionality.

System Under Test (SUT)

Application: Facebook (www.facebook.com)

Features Under Test:

1. Create New Account (Sign Up)
2. Login (Sign In)

Testing Approach

- Black-box testing – Test cases derived from requirements without examining internal code.
- Equivalence Partitioning – Dividing input data into valid and invalid partitions.
- Boundary Value Analysis – Testing at edges of valid ranges.
- Error Guessing – Based on experience and common failure patterns.
- Positive Testing – Valid inputs expecting successful outcomes.
- Negative Testing – Invalid inputs expecting appropriate error messages.

Test Case Format

Each test case is presented with the following fields in descriptive form:

- Test Case ID: A unique identifier
 - Feature: Create Account or Login
 - Test Scenario: What is being tested
 - Test Steps: Step-by-step execution instructions
 - Test Data: Input values used
 - Expected Result: Expected system behavior
 - Test Type:
Positive/Negative/Boundary/Security/Performance/Usability/Compatibility
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Part A: Test Cases for Facebook Create New Account

Feature Description

Facebook's Create New Account requires the following information:

- First name
 - Last name
 - Email address or mobile number
 - New password
 - Birthday (Month, Day, Year)
 - Gender (Female, Male, Custom)
 - (Optional) Preferred pronouns, gender optional display
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Test Case FCAN_001 – Valid account creation with email

Feature: Create New Account

Test Scenario: Valid account creation using email address

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Enter valid first name
4. Enter valid last name
5. Enter valid email address
6. Enter valid password meeting strength requirements
7. Enter valid birthday (Month, Day, Year)
8. Select gender (Female/Male/Custom)
9. Click "Sign Up" button

Test Data:

- First Name: "John"
- Last Name: "Doe"
- Email: "john.doe@example.com"
- Password: "Pass@1234"
- Date of Birth: 15 August 1990
- Gender: Male

Expected Result: Account created successfully. Confirmation email sent to the provided email address. User is redirected to signup confirmation or welcome page.

Test Type: Positive

Test Case FCAN_002 – Valid account creation with mobile number

Feature: Create New Account

Test Scenario: Valid account creation using mobile number instead of email

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Enter valid first name
4. Enter valid last name
5. Enter valid mobile number with country code
6. Enter valid password
7. Enter valid birthday
8. Select gender
9. Click "Sign Up" button

Test Data:

- First Name: "Jane"
- Last Name: "Smith"
- Mobile Number: "+1234567890"
- Password: "Jane@2024"
- Date of Birth: 22 March 1995
- Gender: Female

Expected Result: Account created successfully. SMS confirmation sent to the provided mobile number.

Test Type: Positive

Test Case FCAN_003 – Missing first name

Feature: Create New Account

Test Scenario: User attempts to create account without entering first name

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Leave first name field blank
4. Fill all other fields correctly
5. Click "Sign Up" button

Test Data:

- First Name: (blank)
- Last Name: "Doe"
- Email: "test@example.com"
- Password: "Pass@1234"
- Date of Birth: 15 August 1990
- Gender: Male

Expected Result: Error message displayed: "First name is required" or "What's your name?" Account creation is blocked.

Test Type: Negative

Test Case FCAN_004 – Missing last name

Feature: Create New Account

Test Scenario: User attempts to create account without entering last name

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Enter valid first name
4. Leave last name field blank
5. Fill all other fields correctly
6. Click "Sign Up" button

Test Data:

- First Name: "John"
- Last Name: (blank)
- Email: "test@example.com"
- Password: "Pass@1234"

- Date of Birth: 15 August 1990
- Gender: Male

Expected Result: Error message displayed: "Last name is required" Account creation is blocked.

Test Type: Negative

Test Case FCAN_005 – Invalid email format (missing @ symbol)

Feature: Create New Account

Test Scenario: User enters email address without @ symbol

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Fill all fields correctly
4. Enter email without "@" symbol
5. Click "Sign Up" button

Test Data:

- First Name: "John"
- Last Name: "Doe"
- Email: "johndoe.example.com" (no @ symbol)
- Password: "Pass@1234"
- Date of Birth: 15 August 1990
- Gender: Male

Expected Result: Error message displayed: "Please enter a valid email address" Account creation is blocked.

Test Type: Negative

Test Case FCAN_006 – Invalid email format (missing domain)

Feature: Create New Account

Test Scenario: User enters email with @ but no domain name

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Fill all fields correctly
4. Enter email with @ but no domain

5. Click "Sign Up" button

Test Data:

- Email: "johndoe@.com" (no domain before .com)

Expected Result: Error message displayed: "Please enter a valid email address"

Test Type: Negative

Test Case FCAN_007 – Invalid email format (missing username)

Feature: Create New Account

Test Scenario: User enters email with @ but no username before @

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Fill all fields correctly
4. Enter email with no username before @
5. Click "Sign Up" button

Test Data:

- Email: "@example.com" (no username)

Expected Result: Error message displayed: "Please enter a valid email address"

Test Type: Negative

Test Case FCAN_008 – Email already registered

Feature: Create New Account

Test Scenario: User attempts to create account using an email already associated with an existing Facebook account

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Fill all fields correctly
4. Enter an email address known to already have a Facebook account
5. Click "Sign Up" button

Test Data:

- Email: "alreadyused@gmail.com" (pre-registered email)

Expected Result: Error message displayed: "This email is already connected to an account" OR user is redirected to login page with option to recover password.

Test Type: Negative

Test Case FCAN_009 – Weak password (too short – less than 6 characters)

Feature: Create New Account

Test Scenario: User enters password shorter than minimum required length

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Fill all fields correctly
4. Enter password with fewer than 6 characters
5. Click "Sign Up" button

Test Data:

- Password: "123" (3 characters only)

Expected Result: Error message displayed: "Password is too short" or "Your password must be at least 6 characters"

Test Type: Negative / Boundary

Test Case FCAN_010 – Weak password (no numbers or special characters)

Feature: Create New Account

Test Scenario: User enters password containing only letters with no numbers or special characters

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Fill all fields correctly
4. Enter password with only alphabetic characters
5. Click "Sign Up" button

Test Data:

- Password: "abcdef" (letters only, 6 characters)

Expected Result: Error message displayed: "Password must contain a number or special character"

Test Type: Negative

Test Case FCAN_011 – Weak password (common weak password)

Feature: Create New Account

Test Scenario: User enters commonly used weak password that is easily guessable

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Fill all fields correctly
4. Enter a common weak password like "password", "123456", or "qwerty"
5. Click "Sign Up" button

Test Data:

- Password: "password" or "123456" or "qwerty"

Expected Result: Error message displayed: "Password is too easy to guess. Please choose a stronger password"

Test Type: Negative

Test Case FCAN_012 – Boundary test (minimum valid password length – exactly 6 characters)

Feature: Create New Account

Test Scenario: User enters password with exactly the minimum required length (6 characters) containing a mix of letters and numbers

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Fill all other fields with valid data
4. Enter exactly 6-character password with mix
5. Click "Sign Up" button

Test Data:

- Password: "Pass12" (6 characters with letters and numbers)

Expected Result: Account created successfully (provided all other fields are valid). No

password length error.

Test Type: Boundary Positive

Test Case FCAN_013 – Boundary test (maximum password length)

Feature: Create New Account

Test Scenario: User enters very long password to test maximum character limit

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Fill all other fields with valid data
4. Enter a very long password (e.g., 100+ characters)
5. Click "Sign Up" button

Test Data:

- Password: [100 character string containing letters, numbers, special characters]

Expected Result: System either accepts the long password OR displays appropriate message about maximum length. No crash or unexpected behavior.

Test Type: Boundary

Test Case FCAN_014 – Invalid birthday (user under 13 years old)

Feature: Create New Account

Test Scenario: User enters birthday showing age less than 13 years

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Fill all fields correctly
4. Enter birthday that makes user age under 13
5. Click "Sign Up" button

Test Data:

- Date of Birth: 15 August 2018 (age 6 years)

Expected Result: Error message displayed: "You are not old enough to create a Facebook account" Account creation blocked.

Test Type: Negative / Boundary

Test Case FCAN_015 – Invalid birthday (future date)

Feature: Create New Account

Test Scenario: User enters a date of birth that has not occurred yet

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Fill all fields correctly
4. Enter a future date as birthday
5. Click "Sign Up" button

Test Data:

- Date of Birth: 15 August 2030 (future date)

Expected Result: Error message displayed: "Please enter a valid date of birth"

Test Type: Negative

Test Case FCAN_016 – Invalid birthday (impossible date)

Feature: Create New Account

Test Scenario: User enters an impossible date like February 31st

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Fill all fields correctly
4. Enter an impossible date combination
5. Click "Sign Up" button

Test Data:

- Date of Birth: 31 February 1995 (February has only 28/29 days)

Expected Result: Error message displayed: "Please enter a valid date" or date picker prevents selection of invalid date.

Test Type: Negative

Test Case FCAN_017 – Boundary test (minimum valid age – exactly 13 years)

Feature: Create New Account

Test Scenario: User enters birthday showing age exactly 13 years

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Fill all fields correctly
4. Enter birthday where user turns 13 today or is exactly 13
5. Click "Sign Up" button

Test Data:

- Date of Birth: [Current Year minus 13] on current month and day

Expected Result: Account creation allowed (provided all other fields are valid). No age restriction error.

Test Type: Boundary Positive

Test Case FCAN_018 – Boundary test (maximum valid age – very old user)

Feature: Create New Account

Test Scenario: User enters a very old birthday (e.g., year 1900)

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Fill all fields correctly
4. Enter a very old birth year
5. Click "Sign Up" button

Test Data:

- Date of Birth: 1 January 1900 (age approximately 124+ years)

Expected Result: Account creation allowed (Facebook permits users of all ages above 13). No upper age limit.

Test Type: Boundary

Test Case FCAN_019 – No gender selected

Feature: Create New Account

Test Scenario: User fills all fields but does not select any gender option

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Fill all fields correctly including name, email, password, birthday
4. Do not select any gender (Female, Male, or Custom)
5. Click "Sign Up" button

Test Data:

- Gender: None selected

Expected Result: Error message displayed: "Please select a gender" Account creation blocked until gender is selected.

Test Type: Negative

Test Case FCAN_020 – Custom gender with custom pronoun

Feature: Create New Account

Test Scenario: User selects "Custom" gender and chooses a custom pronoun

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Fill all other fields correctly
4. Select "Custom" as gender
5. Select or enter a custom pronoun (e.g., They/them)
6. Click "Sign Up" button

Test Data:

- Gender: Custom
- Pronoun: "They/them"

Expected Result: Account created successfully. Custom gender and pronoun are saved and displayed appropriately on profile.

Test Type: Positive

Test Case FCAN_021 – Mobile number already registered

Feature: Create New Account

Test Scenario: User attempts to create account using a mobile number already linked

to an existing Facebook account

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Fill all fields correctly
4. Enter a mobile number known to already have a Facebook account
5. Click "Sign Up" button

Test Data:

- Mobile Number: "+1234567890" (pre-registered number)

Expected Result: Error message displayed: "This mobile number is already connected to an account" OR user is redirected to login.

Test Type: Negative

Test Case FCAN_022 – Password confirmation mismatch

Feature: Create New Account

Test Scenario: If Facebook displays a confirm password field, user enters mismatching passwords

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Fill all fields correctly
4. Enter password in password field
5. Enter a different password in confirm password field
6. Click "Sign Up" button

Test Data:

- Password: "Pass@123"
- Confirm Password: "Pass@456" (different)

Expected Result: Error message displayed: "Passwords do not match"

Test Type: Negative

Test Case FCAN_023 – Special characters in name

Feature: Create New Account

Test Scenario: User enters names containing special characters like @, #, \$

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Enter first name with special characters
4. Enter last name with special characters
5. Fill all other fields correctly
6. Click "Sign Up" button

Test Data:

- First Name: "John@"
- Last Name: "Doe#"

Expected Result: Error message displayed: "Please use only letters, spaces, periods, and hyphens"

Test Type: Negative

Test Case FCAN_024 – Extremely long name

Feature: Create New Account

Test Scenario: User enters very long first name or last name (e.g., 255 characters)

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Enter a very long string as first name (255 characters)
4. Fill all other fields correctly
5. Click "Sign Up" button

Test Data:

- First Name: [255 character string of letters]
- Last Name: "Doe"

Expected Result: System either accepts the long name (truncating if necessary) or displays a maximum length message. No crash.

Test Type: Boundary

Test Case FCAN_025 – Spaces only in name

Feature: Create New Account

Test Scenario: User enters only spaces in name fields

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Enter only spaces in first name field
4. Enter only spaces in last name field
5. Fill all other fields correctly
6. Click "Sign Up" button

Test Data:

- First Name: " " (spaces only)
- Last Name: " " (spaces only)

Expected Result: Error message displayed: "Name is required" or "Please enter a valid name"

Test Type: Negative

Test Case FCAN_026 – International mobile number without country code

Feature: Create New Account

Test Scenario: User enters mobile number without country code prefix

Test Steps:

1. Navigate to www.facebook.com
2. Click "Create New Account" button
3. Fill all fields correctly
4. Enter mobile number without + and country code
5. Click "Sign Up" button

Test Data:

- Mobile Number: "1234567890" (no + or country code)

Expected Result: Error message displayed OR system auto-formats with default country code. Clear guidance provided to user.

Test Type: Negative

Test Case FCAN_027 – Browser back button after successful creation

Feature: Create New Account

Test Scenario: After successful account creation, user clicks browser back button

Test Steps:

1. Complete signup successfully with valid data
2. After reaching confirmation/welcome page
3. Click browser back button

Test Data: N/A

Expected Result: User should NOT be able to resubmit the same signup data. Session should be aware that account already exists. Sensitive information should not be exposed in browser history.

Test Type: Security

Test Case FCAN_028 – Network interruption during signup submission

Feature: Create New Account

Test Scenario: User clicks Sign Up button but network disconnects during processing

Test Steps:

1. Fill all signup fields with valid data
2. Click "Sign Up" button
3. Immediately disconnect internet (turn off WiFi/mobile data)
4. Wait 5 seconds
5. Reconnect internet

Test Data: Any valid signup data

Expected Result: Appropriate error message displayed: "Network error. Please try again." No partial account should be created. No data corruption.

Test Type: Performance / Reliability

Test Case FCAN_029 – Signup form persistence on page refresh

Feature: Create New Account

Test Scenario: User fills signup form partially and refreshes the page

Test Steps:

1. Navigate to Create New Account form
2. Fill some fields (e.g., first name, last name, email)
3. Leave other fields empty or partially filled

4. Refresh the browser page (F5 or Ctrl+R)

Test Data: Partially filled form data

Expected Result: Form either clears all fields (recommended for security) OR shows a warning: "Unsaved changes will be lost." No submission occurs.

Test Type: Usability

Test Case FCAN_030 – Accessibility – keyboard navigation

Feature: Create New Account

Test Scenario: User navigates the signup form using only keyboard (Tab key)

Test Steps:

1. Navigate to Create New Account form
2. Use Tab key to move between form fields
3. Verify focus order is logical (top to bottom, left to right)
4. Use Enter key to submit form

Test Data: N/A

Expected Result: All form fields reachable via Tab key. Visual focus indicator visible on each field. Enter key submits form. No keyboard traps.

Test Type: Usability / Accessibility

Part B: Test Cases for Facebook Login

Feature Description

Facebook Login requires:

- Email address or mobile number
 - Password
 - (Optional) "Keep me logged in" checkbox
 - "Forgotten password?" link
 - "Create New Account" link
-

Test Case FLOG_001 – Valid login with email

Feature: Login

Test Scenario: User successfully logs in using valid email and password

Test Steps:

1. Navigate to www.facebook.com
2. Enter valid registered email address in email/phone field
3. Enter corresponding valid password
4. Click "Log In" button

Test Data:

- Email: "valid@example.com"
- Password: "CorrectPass123"

Expected Result: User successfully logged in. Home page/news feed displayed. User sees their profile name in top navigation.

Test Type: Positive

Test Case FLOG_002 – Valid login with mobile number

Feature: Login

Test Scenario: User successfully logs in using valid mobile number and password

Test Steps:

1. Navigate to www.facebook.com
2. Enter valid registered mobile number with country code
3. Enter corresponding valid password
4. Click "Log In" button

Test Data:

- Mobile Number: "+1234567890"
- Password: "CorrectPass123"

Expected Result: User successfully logged in. Home page displayed.

Test Type: Positive

Test Case FLOG_003 – Invalid password for valid email

Feature: Login

Test Scenario: User enters correct email but wrong password

Test Steps:

1. Navigate to www.facebook.com

2. Enter valid registered email address
3. Enter incorrect password
4. Click "Log In" button

Test Data:

- Email: "valid@example.com"
- Password: "WrongPassword"

Expected Result: Error message displayed: "The password you entered is incorrect. Please try again." User remains on login page.

Test Type: Negative

Test Case FLOG_004 – Non-existent email

Feature: Login

Test Scenario: User enters email address not registered on Facebook

Test Steps:

1. Navigate to www.facebook.com
2. Enter email that has no Facebook account
3. Enter any password
4. Click "Log In" button

Test Data:

- Email: "nonexistent@example.com"
- Password: "anything"

Expected Result: Error message displayed: "The email you entered isn't connected to an account. Find your account and log in."

Test Type: Negative

Test Case FLOG_005 – Empty email/phone field

Feature: Login

Test Scenario: User leaves email/phone field blank and enters password

Test Steps:

1. Navigate to www.facebook.com
2. Leave email/phone field empty
3. Enter valid password

4. Click "Log In" button

Test Data:

- Email/Phone: (blank)
- Password: "ValidPass123"

Expected Result: Error message displayed: "Email or mobile number is required" Login blocked.

Test Type: Negative

Test Case FLOG_006 – Empty password field

Feature: Login

Test Scenario: User enters valid email but leaves password field blank

Test Steps:

1. Navigate to www.facebook.com
2. Enter valid registered email
3. Leave password field empty
4. Click "Log In" button

Test Data:

- Email: "valid@example.com"
- Password: (blank)

Expected Result: Error message displayed: "Password is required"

Test Type: Negative

Test Case FLOG_007 – Both fields empty

Feature: Login

Test Scenario: User leaves both email/phone and password fields empty

Test Steps:

1. Navigate to www.facebook.com
2. Leave both fields empty
3. Click "Log In" button

Test Data:

- Both fields empty

Expected Result: Error message displayed: "Email or mobile number is required" and/or "Password is required"

Test Type: Negative

Test Case FLOG_008 – Caps Lock warning

Feature: Login

Test Scenario: User turns on Caps Lock while typing password

Test Steps:

1. Navigate to www.facebook.com
2. Turn on Caps Lock key on keyboard
3. Click into password field
4. Type password
5. Observe if warning appears

Test Data: Caps Lock ON

Expected Result: Warning message displayed near password field: "Caps Lock is on" Helps user avoid incorrect password entry.

Test Type: Usability

Test Case FLOG_009 – Show/Hide password toggle

Feature: Login

Test Scenario: User uses eye icon to reveal hidden password

Test Steps:

1. Navigate to www.facebook.com
2. Enter password in password field (dots/masked)
3. Click the eye icon or "Show" button
4. Observe password visibility
5. Click again to hide

Test Data: Any password

Expected Result: Password becomes visible as plain text when eye icon is clicked. Returns to masked dots when clicked again. No security compromise.

Test Type: Usability

Test Case FLOG_010 – Keep me logged in functionality

Feature: Login

Test Scenario: User checks "Keep me logged in" and expects persistent session

Test Steps:

1. Navigate to www.facebook.com
2. Enter valid credentials
3. Check "Keep me logged in" checkbox
4. Click "Log In"
5. After successful login, close browser completely
6. Reopen browser and navigate to facebook.com

Test Data: Valid credentials with "Keep me logged in" checked

Expected Result: User remains logged in after reopening browser. No login prompt required. Session persists.

Test Type: Positive

Test Case FLOG_011 – Keep me logged in – different browser

Feature: Login

Test Scenario: "Keep me logged in" should not persist across different browsers

Test Steps:

1. Login on Chrome browser with "Keep me logged in" checked
2. Close Chrome
3. Open Firefox browser
4. Navigate to facebook.com

Test Data: Same account on different browser

Expected Result: User is NOT logged in on Firefox. Login screen is displayed. Session is browser-specific.

Test Type: Security

Test Case FLOG_012 – Forgotten password flow

Feature: Login

Test Scenario: User clicks "Forgotten password?" link and initiates recovery

Test Steps:

1. Navigate to www.facebook.com
2. Click "Forgotten password?" link below password field
3. Enter email or phone number associated with account
4. Click "Search" or "Send"
5. Follow recovery instructions

Test Data: Valid registered email or phone

Expected Result: User redirected to password reset page. Recovery options provided (email link, SMS code). Password reset flow works correctly.

Test Type: Positive

Test Case FLOG_013 – Too many failed login attempts

Feature: Login

Test Scenario: User enters wrong password multiple times consecutively

Test Steps:

1. Navigate to www.facebook.com
2. Enter valid email
3. Enter wrong password
4. Click "Log In" (fail)
5. Repeat steps 2-4 for 5 or more times

Test Data: Valid email, repeatedly wrong password

Expected Result: After multiple failures, account is temporarily locked or CAPTCHA is displayed. Message: "Too many failed attempts. Please try again later."

Test Type: Security

Test Case FLOG_014 – Login from new device

Feature: Login

Test Scenario: User logs in from a device/browser not previously used

Test Steps:

1. Use a new device or browser that has never logged into this Facebook account
2. Navigate to facebook.com
3. Enter valid email and password
4. Click "Log In"

Test Data: New unrecognized device

Expected Result: Additional security check triggered. User must verify identity via email code, SMS code, or recognize device from known list.

Test Type: Security

Test Case FLOG_015 – Email case insensitivity

Feature: Login

Test Scenario: User enters email with different case (uppercase/lowercase mix)

Test Steps:

1. Navigate to www.facebook.com
2. Enter registered email in mixed case (e.g., John.Doe@Example.com)
3. Enter correct password
4. Click "Log In"

Test Data:

- Email: "John.Doe@Example.com" (actual: "john.doe@example.com")
- Password: correct

Expected Result: Login successful. Email field is case-insensitive.

Test Type: Positive

Test Case FLOG_016 – Password case sensitivity

Feature: Login

Test Scenario: User enters correct password but with wrong case

Test Steps:

1. Navigate to www.facebook.com
2. Enter valid email
3. Enter correct password but with altered case (uppercase/lowercase changed)
4. Click "Log In"

Test Data:

- Password: "PASSWORD123" (actual: "password123")

Expected Result: Login fails. Password is case-sensitive. Error message: "The password you entered is incorrect."

Test Type: Negative

Test Case FLOG_017 – Trailing spaces in email (auto-trimmed)

Feature: Login

Test Scenario: User enters email with trailing spaces – system should trim

Test Steps:

1. Navigate to www.facebook.com
2. Enter valid email followed by spaces (e.g., "valid@example.com ")
3. Enter correct password
4. Click "Log In"

Test Data:

- Email: "valid@example.com " (with trailing spaces)

Expected Result: Login successful. System automatically trims whitespace from email field.

Test Type: Positive

Test Case FLOG_018 – Trailing spaces in password (not trimmed)

Feature: Login

Test Scenario: User enters password with trailing spaces

Test Steps:

1. Navigate to www.facebook.com
2. Enter valid email
3. Enter correct password followed by a space
4. Click "Log In"

Test Data:

- Password: "CorrectPass123 " (with trailing space)

Expected Result: Login fails because space is treated as part of password. Error message displayed.

Test Type: Negative

Test Case FLOG_019 – SQL injection attempt

Feature: Login

Test Scenario: Attacker attempts SQL injection in email field

Test Steps:

1. Navigate to www.facebook.com
2. Enter SQL injection string in email field
3. Enter any password
4. Click "Log In"

Test Data:

- Email: "" OR '1'='1"
- Password: "anything"

Expected Result: Login fails. No SQL error messages displayed. No unauthorized access granted. System properly sanitizes input.

Test Type: Security

Test Case FLOG_020 – XSS (Cross-Site Scripting) attempt

Feature: Login

Test Scenario: Attacker attempts JavaScript injection in email field

Test Steps:

1. Navigate to www.facebook.com
2. Enter JavaScript code in email field
3. Enter any password
4. Click "Log In"

Test Data:

- Email: "<script>alert('XSS')</script>"
- Password: "anything"

Expected Result: No JavaScript execution. System escapes or rejects malicious input. No alert popup appears.

Test Type: Security

Test Case FLOG_021 – Session hijacking attempt

Feature: Login

Test Scenario: Attacker attempts to reuse session cookie from different environment

Test Steps:

1. Login successfully on Browser A
2. Copy session cookie using developer tools
3. Paste cookie into Browser B (different IP/device)
4. Attempt to access facebook.com on Browser B

Test Data: Session cookie from another browser

Expected Result: Session should be invalidated or additional verification required.
Browser B should NOT be logged in automatically.

Test Type: Security

Test Case FLOG_022 – HTTPS and SSL certificate check

Feature: Login

Test Scenario: Verify login page uses secure connection

Test Steps:

1. Navigate to www.facebook.com
2. Check browser address bar
3. Click on padlock icon

Test Data: N/A

Expected Result: URL must begin with "https://". Valid SSL certificate displayed. No security warnings. Certificate issued to *.facebook.com.

Test Type: Security

Test Case FLOG_023 – Login load time performance

Feature: Login

Test Scenario: Measure time taken from clicking Log In to home page display

Test Steps:

1. Navigate to facebook.com
2. Enter valid credentials
3. Start timer
4. Click "Log In"
5. Stop timer when home page fully loads

Test Data: Valid credentials, good internet connection (10+ Mbps)

Expected Result: Login completes within 3-5 seconds on good connection. User

perceives smooth experience.

Test Type: Performance

Test Case FLOG_024 – Login under poor network (2G/3G simulation)

Feature: Login

Test Scenario: User attempts login on slow network connection

Test Steps:

1. Use browser developer tools to throttle network to 2G or 3G speed
2. Navigate to facebook.com
3. Enter valid credentials
4. Click "Log In"
5. Observe behavior

Test Data: Valid credentials, throttled network

Expected Result: Loading indicators displayed. Timeout message if request takes too long. No partial or corrupted state. User can retry.

Test Type: Performance

Test Case FLOG_025 – Concurrent login same account

Feature: Login

Test Scenario: User logs into same account on two different devices simultaneously

Test Steps:

1. Login on Device A (laptop)
2. Login on Device B (mobile phone) with same credentials
3. Observe both devices

Test Data: Same account credentials

Expected Result: Both sessions remain active. User may see notification: "You're logged in on [Device A]." Option to logout from other devices available in settings.

Test Type: Functional

Test Case FLOG_026 – Logout functionality

Feature: Login

Test Scenario: User logs out and session is terminated

Test Steps:

1. Login successfully
2. Click profile

Conclusion:

This assignment successfully designed **60 comprehensive test cases** for Facebook's two most critical user-facing features: **Create New Account (30 test cases)** and **Login (30 test cases)** . The test cases were developed using established black-box testing techniques including equivalence partitioning, boundary value analysis, error guessing, and security testing principles.